

Qatar-1b

R-2755

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_160719 · created 2026-09-02T23:56:27+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457588.80649 +/- 0.00083 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.183 +/- 0.056
Transit depth (Rp/Rs)^2	3.36 +/- 2.05 [%]
Orbital Inclination (inc)	79.09 +/- 2.92
Airmass coefficient 1 (a1)	6.83 +/- 1.04
Airmass coefficient 2 (a2)	-1.53 +/- 1.02
Scatter in the residuals of the lightcurve fit is	15.59 %
Best Comparison Star	#3 - [306, 156]
Optimal Aperture	1.68
Optimal Annulus	11.1
Transit Duration (day)	0.023 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.183 ± 0.056 vs 0.1463 (deviation 0.66σ)
Duration fit vs published	0.023 d vs 0.0692 d (deviation 1.71σ)
Mid-time O–C	0.1 min
Depth SNR	3.3
Residual scatter	15.59 %

Light curve

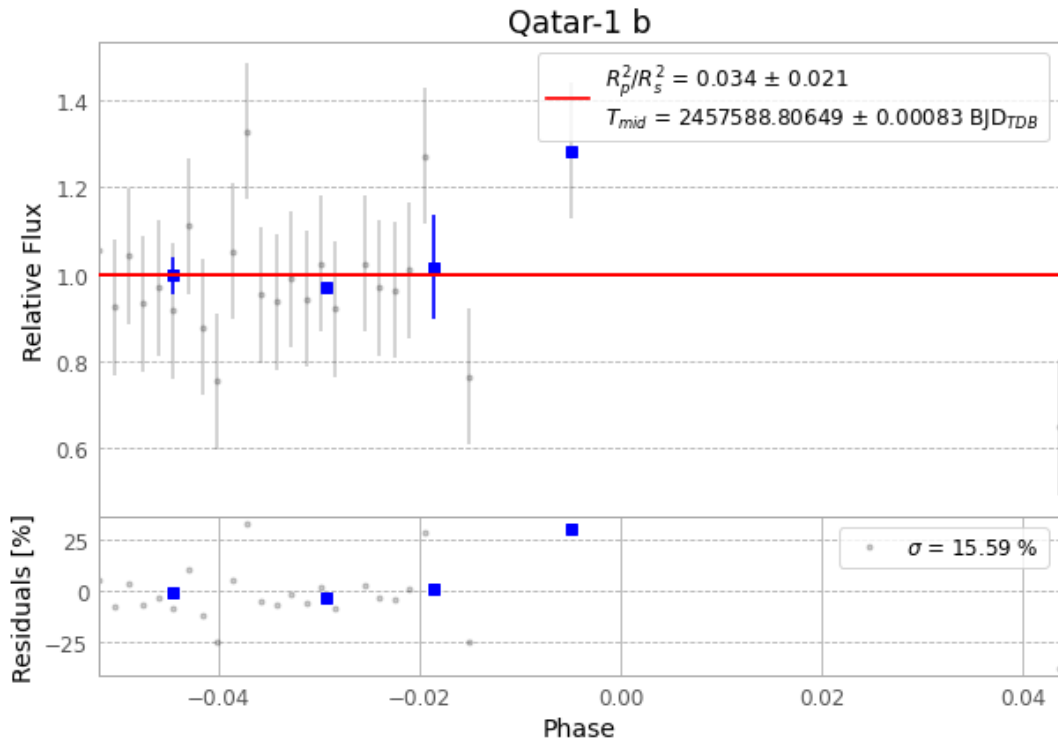


Figure 1 — FinalLightCurve_Qatar-1 b_2016-07-18.png (SHA-256 0761f799feb4f6c8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [258, 212], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2bcd7eb46488ee6a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

73 FITS frames (48 MB), Qatar-1160719053312.FITS → Qatar-1160719090912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-160719.log (3975b64c1150...), FinalLightCurve_Qatar-1 b_2016-07-18.png (0761f799feb4...), FinalLightCurve_Qatar-1 b_2016-07-18.csv (d4fe8225c1ae...)

Compute resources

Wall time 303.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 23:56Z.